

A steady 2D finite-volume SIMPLE solver for incompressible Navier–Stokes flow

Architecture, verification, and lid-driven cavity validation

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September 2026

Abstract

This paper documents the frozen steady version of a compact C++ solver for the two-dimensional incompressible Navier–Stokes equations. The code uses a cell-centered collocated finite-volume discretization, a separately assembled pressure-correction equation, and the SIMPLE fixed-point algorithm. Diffusion and convection are verified first on scalar transport problems, then the complete pressure–velocity coupling is assessed with lid-driven cavity calculations. The $\text{Re}=100$ cavity on a 32×32 grid reproduces the Ghia centerlines with L_2 errors of 1.52×10^{-3} for u and 4.08×10^{-3} for v . The same frozen centered formulation does not reach the stopping criteria at $\text{Re}=400$ or $\text{Re}=1000$ within 5000 iterations. This limitation is measured through residual histories, local cell Peclet numbers, grid probes, and linear-solver diagnostics. The result is a reproducible and well-tested steady baseline, but not yet a robust production solver at all target Reynolds numbers.

1 Scope and governing equations

The objective of this implementation is a small, readable, and performant steady incompressible flow solver that can be extended without entangling the physical operators. The frozen baseline solves

$$\nabla \cdot \mathbf{u} = 0, \quad \rho \nabla \cdot (\mathbf{u}\mathbf{u}) = -\nabla p + \nabla \cdot (\mu \nabla \mathbf{u}), \quad (1)$$

with constant density and viscosity in the reported cases. There is no body force. The main validation problem is the unit-square lid-driven cavity with no-slip walls and a unit tangential velocity on the top wall; its Reynolds number is $\text{Re} = \rho U_{\text{lid}} L / \mu$.

The solver is intentionally organized around data and operators rather than around a large class hierarchy. The mesh owns Cartesian geometry and row-major cell indexing, the transport routines assemble sparse momentum systems, ‘Simple.cpp’ constructs the pressure correction and diagnostics, and the linear solver is isolated in ‘Solver.cpp’. Output is written as plain field and diagnostic files, so a run can be reproduced from a text input file. This separation leaves pressure, convection, and diffusion as distinct numerical components while retaining a minimal steady executable.

2 Finite-volume discretization

2.1 Cell-centered transport

The mesh is collocated and cell-centered. A cell $P = (i, j)$ has index $k = i + N_x j$ and receives one scalar value for each of u , v , and p . Integrating a generic steady transport equation over the cell gives

$$\sum_f F_f \phi_f = \sum_f D_f (\phi_N - \phi_P) + S_P, \quad F_f = \rho (\mathbf{u}_f \cdot \mathbf{n}_f) S_f, \quad D_f = \frac{\Gamma_f S_f}{d_{PN}}. \quad (2)$$

Diffusive fluxes use a two-point face gradient. Centered convection uses $\phi_f = (\phi_P + \phi_N)/2$, while upwind convection selects the upstream cell according to the sign of F_f . Face values on Dirichlet boundaries are the prescribed values; the existing half-cell closure is retained.

The sparse matrices are assembled directly from local coefficients using Eigen triplets. No global interpolation, divergence, or gradient matrices are stored. This keeps memory proportional to the five-point stencil and makes the operator boundary between assembly and solution explicit. In the TP2 matrix regression, a $N_x = 5$ centered case with $U = \Gamma = 0.1$ recovers the expected interior off-diagonals $a_E = -0.45$ and $a_W = -0.55$ after diffusion. The tests also cover reversed flow direction, variable-sign face velocities, the single-cell limit, and exact equivalence with diffusion when convection is zero.

Table 1: Main components of the frozen implementation.

Component	Role in the calculation
Mesh	Uniform Cartesian cell geometry, face areas, distances, and neighbors.
Transport assembly	Diffusion plus centered or first-order upwind convection, with boundary closure.
Momentum	Separate u and v equations with the current pressure source.
SIMPLE	Momentum prediction, pressure correction, face-flux correction, relaxation, and stopping tests.
Solver and output	Sparse BiCGSTAB/ILUT with SparseLU fallback; CSV-like fields and iteration diagnostics.

2.2 Pressure-velocity coupling

SIMPLE is applied to the same cell-centered fields. For the current pressure p^k , the momentum predictor is written schematically as

$$A_u u^* = b_u - G_x p^k, \quad A_v v^* = b_v - G_y p^k. \quad (3)$$

The pressure-correction equation is assembled from the diagonal inverse of the momentum matrix. For an internal east face, for example,

$$u'_e = d_{u,e}(p'_P - p'_E), \quad d_{u,e} = \frac{1}{2} \left(\frac{\alpha_u \Delta y}{a_{P,u}} + \frac{\alpha_u \Delta y}{a_{E,u}} \right), \quad (4)$$

and analogous north-face terms give a five-point pressure-correction system $A_p p' = -r_m$. The first pressure-correction row is fixed to zero to remove the Neumann null space. After solving,

$$p^{k+1} = p^k + \alpha_p^{\text{eff}} p', \quad u^{k+1} = u^* + u', \quad v^{k+1} = v^* + v'. \quad (5)$$

The cell-centered face reconstruction contains the pressure-difference term used by the pressure-correction flux. It is the continuity representation used by SIMPLE and suppresses the direct pressure-velocity decoupling that would result from plain arithmetic interpolation alone.

The pressure is re-gauged after every iteration by subtracting the value in the reference cell. On a 32×32 square grid, the reproducible cavity settings are $\alpha_u = \alpha_v = 1$, ten inner momentum iterations, $\alpha_p = 0.1$, and a pressure-relaxation cap factor of 0.1. The effective pressure relaxation is therefore $\alpha_p^{\text{eff}} = \min(\alpha_p, 0.1 h_{\min}) = 0.003125$. The stopping criteria require both momentum residuals and both face and reconstructed cell divergence norms to be below 10^{-6} .

2.3 Linear algebra and diagnostics

Each sparse system is first attempted with BiCGSTAB and an ILUT preconditioner, using tolerance 10^{-10} and 2000 iterations. A failed or inaccurate iterative result falls back to SparseLU. This choice is simple and robust enough for the small development cases, but the counters are recorded because repeated fallbacks affect runtime and can hide conditioning problems.

Two continuity diagnostics are kept separate. The corrected face fluxes are the conservative SIMPLE fluxes; their divergence is nearly machine precision when the pressure-correction equation is solved accurately. A second divergence is reconstructed from cell velocities and pressure, using the collocated face formula. Both are reported, along with the momentum residuals, field updates, local cell Peclet number, matrix diagonal, and diagonal dominance margin.

3 Verification of the building blocks

The frozen steady regression suite contains scalar convection-diffusion, momentum without pressure, pressure-driven channel, centered-matrix, input-parser, pressure-correction consistency, and small SIMPLE cavity tests. All seven steady tests pass in the current build. They are deliberately small: their role is to detect coefficient, sign, boundary, gauge, and conservation regressions before expensive cavity runs.

The quantitative mesh study with a known solution is the one-dimensional steady transport problem at $\text{Pe} = 10$,

$$\phi(x) = \frac{\exp(\text{Pe}x/L) - 1}{\exp(\text{Pe}) - 1}, \quad \phi(0) = 0, \quad \phi(L) = 1. \quad (6)$$

The measured asymptotic orders are 1.98 for centered convection and 0.98 for upwind in the L_2 norm, as shown in Fig. 1. This result confirms the expected spatial order of both transport choices and provides a stronger verification than a visual flow-field comparison.

The pressure-free momentum test verifies that the u and v systems reduce to the transport operator when the pressure gradient is zero. The channel case then exercises a non-zero pressure source and wall boundaries. Finally, the small 4×4 cavity regression converges with an L_∞ face divergence of 9.9×10^{-7} . Its velocity field and the coarse-grid recirculation sweep are shown in Fig. 2. The latter is a development diagnostic, not a claim of grid-independent vortex location.

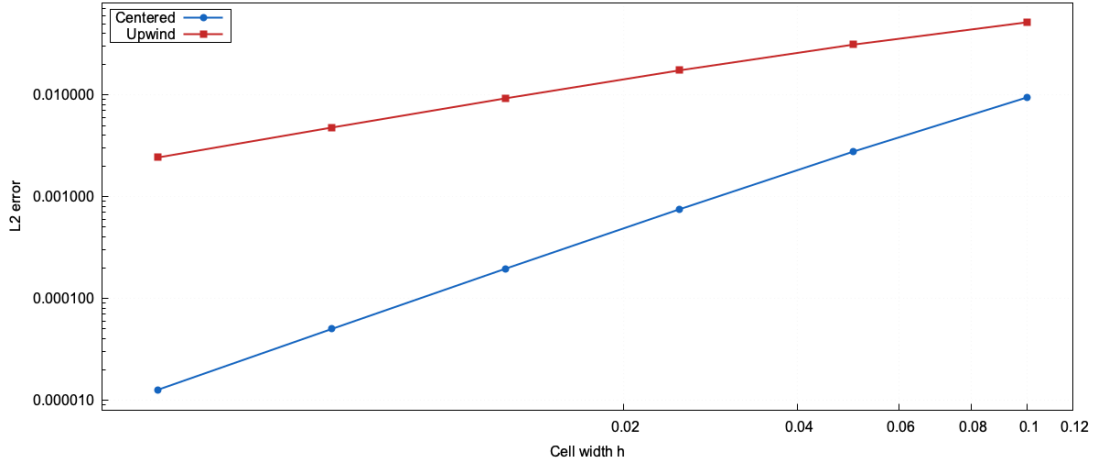


Figure 1: Scalar transport mesh convergence at $Pe = 10$. Centered convection approaches second order and first-order upwind approaches first order.

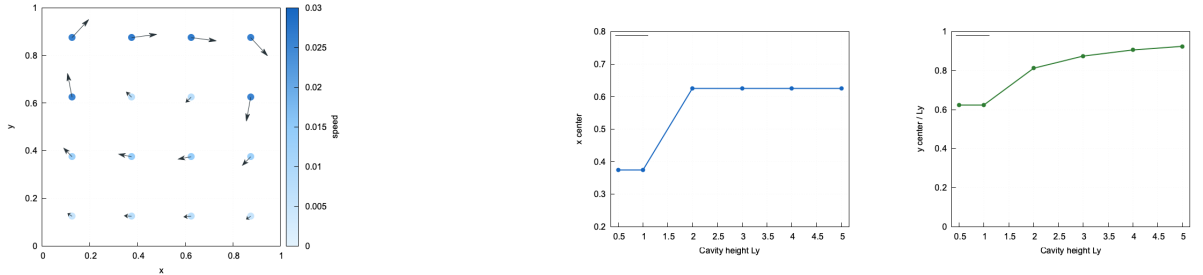


Figure 2: Frozen SIMPLE development results: centered 4×4 cavity field (left) and primary recirculation estimates for varying cavity height (right).

4 Lid-driven cavity validation

4.1 Re=100 baseline

The canonical validation uses $\rho = 1$, $L_x = L_y = 1$, $U_{\text{lid}} = 1$, and $\mu = 1/\text{Re}$ on a 32×32 mesh. Centerline profiles are compared to the tabulated data of Ghia, Ghia, and Shin. At $\text{Re} = 100$, the run converges after 4928 SIMPLE iterations. The conservative face divergence is 3.22×10^{-15} and the reconstructed cell-divergence L_∞ norm is 9.98×10^{-7} .

The $\text{Re}=100$ velocity errors are $\|e_u\|_2 = 1.52 \times 10^{-3}$, $\|e_u\|_\infty = 3.11 \times 10^{-3}$, $\|e_v\|_2 = 4.08 \times 10^{-3}$, and $\|e_v\|_\infty = 9.15 \times 10^{-3}$. The primary vortex estimate is $(0.6151, 0.7374)$, compared with $(0.6172, 0.7344)$ in the reference table. These errors are consistent with a useful coarse-grid baseline. They do not, by themselves, establish spatial convergence: the requested 32/64/128 steady sequence was not completed successfully, and `grid_convergence.csv` contains no valid coarse/fine pair.

4.2 What fails at higher Reynolds number

The same centered discretization reaches the iteration limit at $\text{Re}=400$ and $\text{Re}=1000$. The issue is not a catastrophic blow-up: velocities remain bounded, the face flux balance remains small, and no non-positive momentum diagonal is observed. Instead, the reconstructed cell divergence and field updates settle into oscillatory plateaus. Figure 4 makes the distinction between an accurate pressure-correction solve and a converged nonlinear fixed point visible.

5 Convergence audit and convection control

The audit links the failure to the centered cell Peclet number rather than to a single relaxation setting. At the final 32×32 iterates, the maximum local $Pe_\Delta = |u_f|\Delta/\nu$ is 2.82, 8.77, and 22.70 for $\text{Re}=100, 400$, and 1000, respectively. The 95th percentiles are 1.20, 3.90, and 4.26. Refinement reduces these values in one-iteration probes, but it does not make the nonlinear fixed point converge inside the tested budget. The trend is summarized in Fig. 5.

For the centered momentum matrices, the minimum diagonal remains positive, but the minimum diagonal-dominance margin becomes negative: -2.43×10^{-3} , -1.75×10^{-2} , and -2.14×10^{-2} at the three Reynolds numbers. This is consistent with centered convection becoming increasingly difficult to bound at large cell Peclet number. Relaxation screening over $\alpha_u = \alpha_v \in \{0.3, 0.7, 1.0\}$ and requested pressure relaxation $\alpha_p \in \{0.001, 0.003, 0.01\}$ did not produce

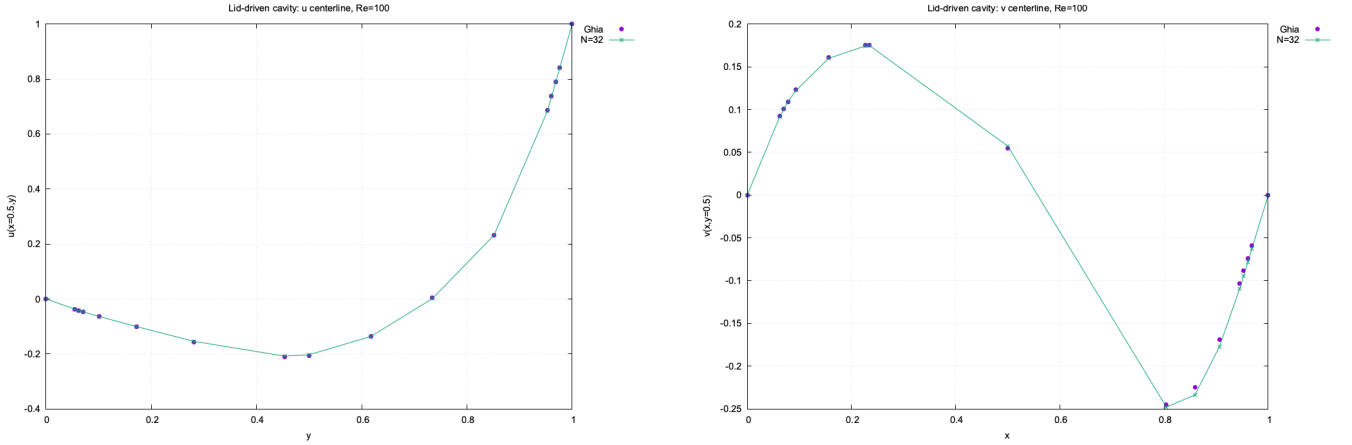


Figure 3: Re=100 centerline validation on 32×32 : vertical u profile (left) and horizontal v profile (right), compared with Ghia et al.

Table 2: Lid-driven cavity results on 32×32 . A non-converged run is reported as a diagnostic and is not used as a validated reference state.

Quantity	Re = 100	Re = 400	Re = 1000
Converged	yes	no	no
SIMPLE iterations	4928	5000	5000
u residual	3.49×10^{-11}	1.45×10^{-4}	6.32×10^{-5}
v residual	3.53×10^{-11}	1.01×10^{-4}	5.98×10^{-5}
Face div. L_∞	3.22×10^{-15}	1.79×10^{-13}	1.54×10^{-13}
Cell div. L_∞	9.98×10^{-7}	2.91×10^{-1}	7.29×10^{-1}
Vortex (x, y)	(0.6151, 0.7374)	—	—
Kinetic energy	3.290×10^{-2}	—	—

a converged Re=400 or Re=1000 case. A no-cap pressure control also failed; removing the cap is therefore not a solution.

The linear algebra is a secondary performance limitation. In the Re=100, 400, and 1000 baseline runs, the code recorded respectively 34,524, 35,122, and 35,710 BiCGSTAB solves, with 24,465, 18,567, and 18,637 SparseLU fallbacks. The fallback count explains why the nominally small 32×32 cases take about two minutes. A symmetric pressure solver or a better preconditioner is a future optimization, but the audit does not justify calling the current nonlinear failure a purely linear-solver failure.

The first-order upwind scheme is retained as a control rather than silently used as a production fallback. It adds numerical diffusion, improves matrix conditioning, and reduces the short-run divergence at Re=400 and Re=1000, but it also changes the physical discretization and does not converge under the same 1000-iteration control budget. The earlier transport sweep in Fig. 6 shows the expected centered/upwind trade-off.

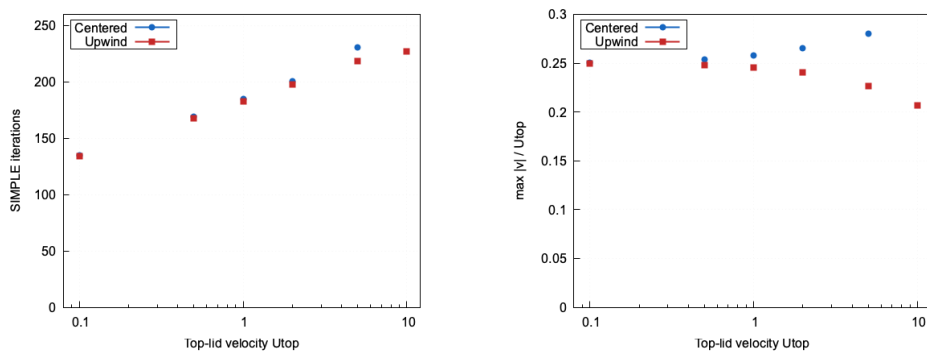


Figure 6: Centered versus first-order upwind convection on the development cavity. Upwind is more diffusive and remains bounded at stronger transport, but is not silently substituted for the centered reference scheme.

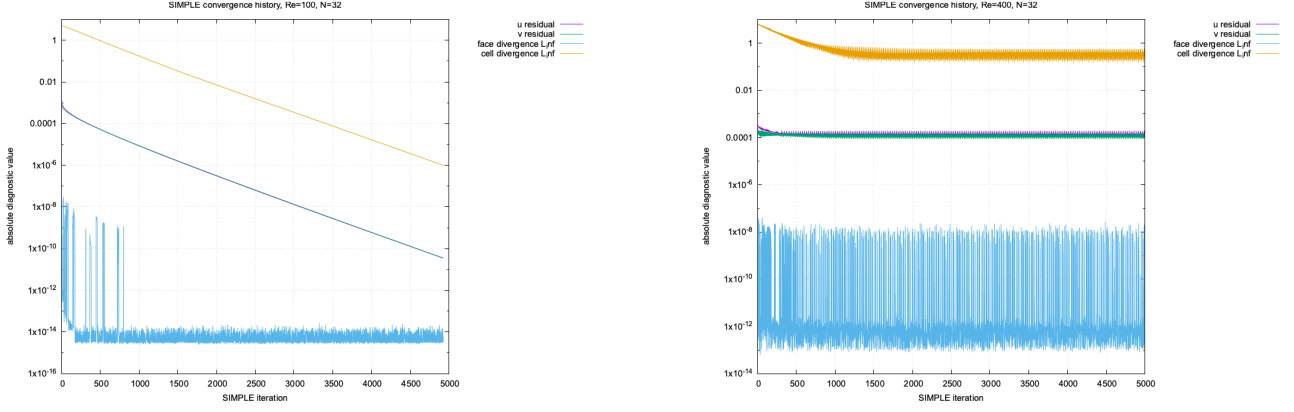


Figure 4: SIMPLE histories on 32×32 : Re=100 converges slowly (left), whereas Re=400 reaches an oscillatory residual and cell-divergence plateau (right). The Re=1000 history has the same behavior with a larger plateau.

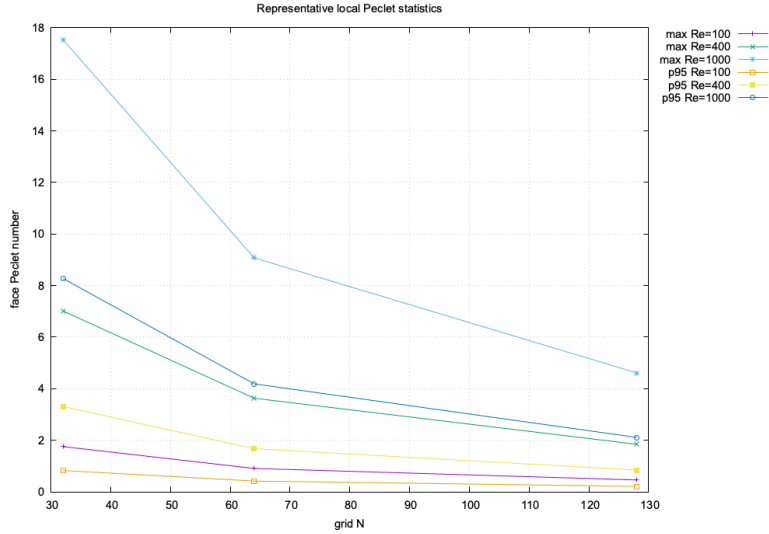


Figure 5: Local face Peclet statistics from grid probes. Refinement lowers the cell Peclet number, but this alone did not restore steady SIMPLE convergence.

6 Reproducibility, limitations, and conclusion

The frozen baseline is reproduced from the repository root with

```
cmake -S . -B build -DBUILD_TESTING=ON
cmake --build build --parallel
ctest --test-dir build --output-on-failure
./build/cfd_solver input/cavity.txt build/cavity_output
```

The input file controls mesh size, geometry, viscosity, boundary conditions, convection scheme, relaxation, and stopping limits. The run reports both the requested and effective pressure relaxation and writes u , v , and p cell fields together with diagnostics. Benchmark figures in this paper are generated from the committed validation data under ‘validation/’.

The completed part of the project is therefore precise: diffusion and convection have unit-style matrix and known-solution verification; pressure correction conserves the corrected face flux; SIMPLE is reproducibly converged for the Re=100 cavity and for the small teaching cases; and the centerline and vortex diagnostics agree with the Re=100 reference. The remaining limitation is equally precise: the frozen centered collocated SIMPLE iteration is not a validated steady method at Re=400 or Re=1000, and no grid-independent cavity result is claimed. The next scientifically useful step is to improve or replace the steady pressure-velocity iteration while keeping the existing operator tests and the Re=100 baseline as regression anchors.

References

- [1] U. Ghia, K. N. Ghia, and C. T. Shin, “High-Re solutions for incompressible flow using the Navier–Stokes equations and a multigrid method,” *Journal of Computational Physics*, 48(3), 387–411, 1982.
- [2] O. Botella and R. Peyret, “Benchmark spectral results on the lid-driven cavity flow,” *Computers & Fluids*, 27(4), 421–433, 1998.